MITOSYNC: PROGRAMMABLE DIVISION SUPERVISION FOR DIVISION-AWARE CELL TRACKING

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ABSTRACT

Sparse sampling and limited annotations make mother–daughter association difficult. MitoSync scores each mother–daughter triplet without depending on daughter order, then selects divisions and continuations jointly. Its optional supervision pipeline supplies synthetic training videos and aligns division labels with detected cells. On two retrospectively evaluated Cell Tracking Challenge sequences, real-only joint association exceeds the Trackastra graph reference by 5.24 and 7.95 sparse division F1 points. On 36 external LFCT fields, the frozen real-only joint model gains 4.05 sparse event F1 points; native-rate gains are not established. With the full tracker fixed, synthetic data add 0.79–1.70 points in three settings; matched controls establish no consistent advantage over painting or real-only training. Joint association supplies the main tracking gain and transfers to new videos under sparse sampling.

***Index Terms*—** cell division, cell tracking, synthetic supervision, video diffusion, lineage

1. INTRODUCTION

Missed divisions and false daughter assignments corrupt lineages. Divisions are rare in tracker training data [1, 2]: fewer than 7 per 1000 cell-frames in our PhC-C2DL-PSC (PSC) training split. Sparse sampling separates mother and daughter observations in time. Plausible links can assign daughters from different lineages to one mother.

Synthetic videos supply examples, but programmed divisions may appear late or be absent. MitoSync scores daughter compatibility without depending on daughter order. Joint decoding combines this evidence with Trackastra link scores to select divisions and continuations. MitoSync includes the tracker and an optional M1–M3 supervision pipeline (Fig. 1). Controls separate association, appearance, label and sampling effects.

2. RELATED WORK

Cell sequences can be simulated [3, 4] or generated by learned models [5–9]. SynCellFactory (SCF) [10] combines fitted statistics, ControlNet and label refinement; ASIST [11] trains segmentation and tracking with simulations. LapTrack [12], MAGIK [2] and Trackastra [1] link detected cells. Zyss et al. [13] classify mother–two-daughter triplets using three-frame image and geometry features, select consistent divisions, then optimize tracks. We instead couple an exchange-invariant representation to Trackastra through a child-set utility that jointly selects competing divisions and continuations.

3. MITOSYNC

3.1. M1: programming cell divisions

M1 uses training-fitted statistics for lognormal cycles, growth, rounding and motion to generate 17-frame identity masks $L_{0:T}$ and lineage links. Initial ages follow the distribution in an exponentially growing population [14], avoiding synchronized divisions. The mother long axis defines the preferred division axis [15]. Division-axis deviation ψ and daughter separation d follow

$$p(\psi) \propto \exp\{\kappa \cos(2\psi)\}, \qquad d(k) = d_\infty - (d_\infty - d_0)e^{-k/\tau}. \quad (1)$$

Here k counts frames after division. Training contact statistics guide placement, non-overlap and cell entry.

3.2. M2: phase scheduling and video generation

We adapt Wan2.1-VACE-1.3B [16, 17] using rank-32 low-rank adaptation (LoRA) [18] on its 15 VACE blocks. A latent flow-matching loss updates only LoRA parameters, using 113 PSC or 90 Fluo-N2DL-HeLa (Fluo) training windows. Controls encode instance boundaries, separation lines and distance transforms; a real training frame supplies appearance. Generation uses 30 steps and guidance scale 5 (Fig. 2).

The causal variational autoencoder (VAE) encodes frame 0 alone, then groups of four frames. Training calibration selects two birth phases $\phi(t_b) = (t_b - 1) \bmod 4$ with the highest on-time detection rates: $K^* = \{1, 2\}$ for PSC and $\{0, 2\}$ for Fluo. M2 shifts births to the nearest selected phase.

3.3. M3: aligning labels with generated frames

M3 matches frozen Cellpose-SAM [19] detections one-to-one with programmed masks at intersection over union (IoU)$\geq .5$. For a programmed birth t_b, the detected birth is

$$t^* = \min\{t \geq t_b : \text{both daughters match at } t \text{ and } t+1\}. \quad (2)$$

M3 keeps the birth when $t^* = t_b$ and re-times it to t^* when delayed. If no match exists, it removes the split and assigns the daughter regions to the mother identity. This checks detector consistency; it does not independently establish ancestry. On PSC, M3 keeps 1085 of 6047 divisions on time (18%), re-times 2435 and drops 2527. The same rule detects 63% of real training divisions on time.

Acceptance requires 80% matched divisions and PSC instance F1 $\geq .6986$ (.05 below the calibration median). After two retries,

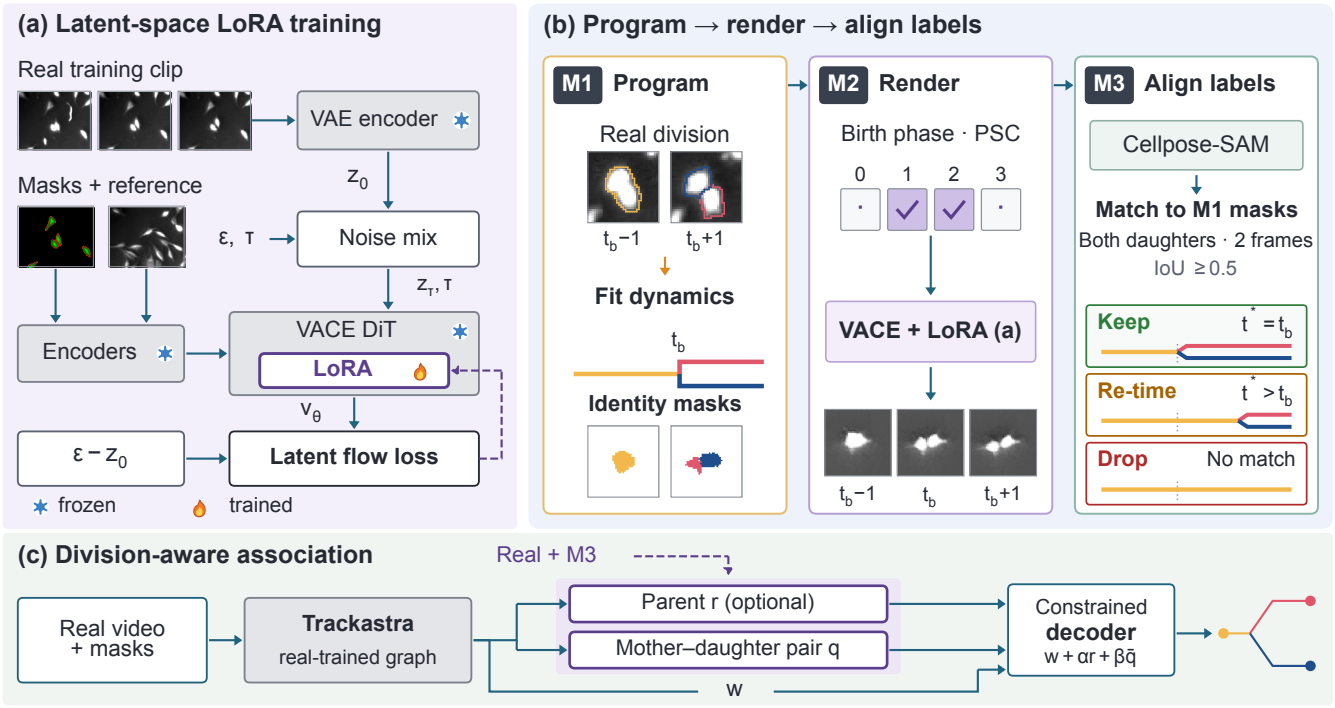


Fig. 1. MitoSync: (a) latent LoRA adaptation; (b) division programming, rendering and label alignment; (c) joint association. Dashed arrows carry training inputs/gradients; the direct path carries Trackastra score w. Yellow/pink/blue denote mother/daughters; τ is diffusion time.

fallback retains the highest-match-rate attempt passing the instance gate (176/191 PSC clips). Retained events satisfy Eq. (2).

3.4. MitoSync tracker

Candidate links and decoding. Real-trained Trackastra [1] scores consecutive links $p \rightarrow c$ with w_{pc}. Native-rate training adds recombined-cell frames; sparse-rate training adds eight subsampled views of the training split. Scores r_{pc} and $q_p(a, b)$ refine links and mother–daughter triplets.

Each parent p selects one child set S; each child has at most one parent. With $g_{pc} = w_{pc} + \alpha r_{pc}$, the utility is

$$U_p(S) = \begin{cases} 0, & S = \emptyset, \\ \sum_{c \in S} g_{pc} + \beta \bar{q}_p(S) + .75(2 - |S|), & 1 \le |S| \le 3. \end{cases} \tag{3}$$

Integer programming maximizes total utility across parents: one child represents continuation and two represent division. Setting $\alpha = \beta = 0$ gives the Trackastra graph reference in Table 1. For up to two children, this is the Trackastra integer linear programming (ILP) objective with `deepcell_gt` costs (.75). With learned scores, parents linked in the graph reference must retain a child. This continuity constraint uses predictions, not test lineage labels. The three-child limit accommodates annotations, not three-way mitosis.

Mother–daughter pair head. A frozen, real-trained ResNet-18 division classifier encodes three-frame cell crops as 256-dimensional vectors, projected to 32 dimensions. At 8×, crops use $t-8, t, t+8$, repeating endpoint frames; padding is flagged. A multilayer perceptron (MLP) combines the mother vector, daughter mean, absolute difference and product, and comparisons between mother and mean daughter features. It also uses 25 geometry/context values for area balance, separation, motion, density, frame gap and boundaries. Both visual and geometry inputs are symmetric; the clipped logit $q_p(a, b) \in [-4, 4]$ is invariant to daughter swapping. Training uses balanced binary cross-entropy with hard negatives from the four nearest candidate children. In Eq. (3), $\bar{q}_p(S)$ averages unordered pair scores, avoiding extra terms for larger sets; it is 0 when $|S| < 2$.

Validation selects $\beta = .125$ for PSC at 1× and .25 at 8× from $\{0, .125, .25\}$. Fluo at 8× uses the PSC weight; its native validation split, containing five divisions, selects .5 from $\{0, .125, .25, .5\}$.

Optional parent head. An MLP scores the ten nearest parents versus entry using images and geometry (9/15 values at 1×/8×). Clipped log odds give r_{pc}. PSC validation selects $\alpha = .25$ for both datasets and rates. Native training uses gap 1; sparse training balances gaps 1/2/4/8. Setting $\alpha = 0$ gives the pair-only model; Full retains both heads for controlled comparisons.

Training. Both heads use final checkpoints after 6000 AdamW updates (batch 64). R uses real data; RG samples real and synthetic data equally. Pair batches balance gaps 1/2/4/8 and binary labels. Five seeds vary initialization, sampling and Trackastra checkpoints; corpora and encoders remain fixed.

4. EXPERIMENTS AND RESULTS

4.1. Experimental setup

Data. PSC and Fluo each provide two annotated sequences [20, 21]. Video 02 provides training frames 0–254 (PSC)/0–77 (Fluo) and validation tails of 45/14 frames for all components. Evaluation uses video 01: PSC (300 frames, 569 divisions) and Fluo (92 frames, 94 divisions). CTC methods share silver-truth masks to compare association under common detections.

Table 1. Video 01 division F1 (BC(0)) and complete tracks (CT), mean±SD; bold maxima. Graph: $\alpha = \beta = 0$; Full/Pair: with/without parent head; +synthetic: M1–M3 data. Shared detections; greedy may omit isolated cells. MAGIK: uniform here, balanced in Sec. 4.4.

	PSC 1×		PSC 8×		Fluo 1×		Fluo 8×	
Method	F1	CT	F1	CT	F1	CT	F1	CT
LapTrack	.730	.556	.589	.518	.856	.764	.760	.683
MAGIK (uniform)	.770±.005	.536±.007	.454±.019	.377±.008	.710±.019	.608±.019	.616±.035	.512±.024
+ synthetic	.797±.010	.571±.011	.472±.026	.392±.014	.729±.063	.642±.036	.500±.070	.456±.050
Trackastra greedy	.837±.007	.651±.010	.696±.007	.592±.007	.943±.009	.861±.015	.854±.024	.787±.021
Trackastra graph	.857±.008	.660±.008	.696±.007	.594±.010	.947±.010	.871±.017	.830±.038	.767±.032
Full (real only)	.862±.006	.671±.006	.741±.003	.669±.004	**.954**±.003	**.884**±.005	.899±.008	.839±.005
Full + SCF-adapted	.857±.009	.665±.009	.731±.005	.660±.004	.946±.007	.880±.005	.891±.006	.831±.003
Full + synthetic	.879±.003	.679±.002	.749±.005	**.674**±.004	.950±.002	**.884**±.006	.909±.006	.846±.003
Pair (real only)	.874±.007	.670±.009	.749±.004	.656±.007	**.954**±.003	**.884**±.005	.909±.007	.848±.004
Pair + synthetic	**.888**±.006	**.682**±.004	**.755**±.003	.667±.004	.950±.002	.880±.005	**.913**±.007	**.854**±.004

Table 2. Component F1 over five seeds; bold marks maxima. No pair: no daughter scores; Geometry pair: zeroed image embeddings; Full: both heads. R/RG: real/real+synthetic.

		PSC 01		Fluo 01	
Stage	Data	1×	8×	1×	8×
No pair	R	.838±.008	.693±.006	.909±.011	.820±.015
	RG	.866±.006	.701±.009	.928±.006	.859±.015
Geometry pair	R	.846±.006	.724±.004	.950±.003	.886±.011
	RG	.872±.004	.737±.006	.950±.002	.898±.007
Full	R	.862±.006	.741±.003	**.954**±.003	.899±.008
	RG	**.879**±.003	**.749**±.005	.950±.002	**.909**±.006

MitoSync produces 191 accepted PSC clips and 200 Fluo clips. Their tracking labels contain 3520 and 3554 matched binary divisions, versus 260 and 202 real training divisions. Fluo includes frame-0 mothers; excluding them for classification gives 3424/199 without changing the corpus. Generation uses 584 PSC and 352 Fluo attempts including retries.

Evaluation. Division F1 measures branching correctness with zero temporal tolerance (BC(0)) [20, 22]; CT measures complete-track recovery [20]. We evaluate native frames (1×) and every eighth frame (8×). Each offset retains frames $o + 8j$, removes unobserved tracks and keeps only surviving parent links, without ancestor substitution. Divisions require a mother and at least two daughters. Eight correlated offsets are averaged per seed. Seed variation does not measure cross-video uncertainty. Five-seed comparisons use paired 95% t intervals (four degrees of freedom) and stated Holm corrections; differences are percentage points. LFCT comparisons, identity checks and uncertainty masking are post hoc. Video 01 informed development and the decision to reselect the Fluo native weight on validation; evaluation is retrospective and exploratory. The repository lists the comparisons in each correction family.

Baselines. LapTrack [12] uses parameters fitted on all of video 02. We train MAGIK [2] on geometry inputs, with and without MitoSync data. SCF-adapted uses released SCF code and schedules [10], with moment-matched training motion statistics. SCF retains simulated-position mask correction but uses generic Cellpose-SAM instead of fine-tuned Cellpose (199/200 programs per dataset). Table 1 uses the output of each pipeline; later controls match event budgets. LapTrack is deterministic; MAGIK uses three seeds and other trackers five.

4.2. Comparison with other trackers

Under sparse sampling (Table 1), full RG exceeds the Trackastra graph by 5.25 F1 points on PSC (95% interval [4.67,5.82]) and 7.93 on Fluo ([3.52,12.34]). Full RG also exceeds the official greedy linker; comparisons include learned scores and continuity constraints. Sparse CT gains are 7.96/7.92 PSC/Fluo points. Native F1 improves clearly only on PSC; Fluo F1/CT intervals include zero. Sparse Pair R gains persist at BC(1)/BC(2): 5.12/5.15 PSC points and 7.92/7.92 Fluo points (tolerances in observed frames). Removing continuity changes sparse Pair R/RG F1 by at most 0.11 points; native PSC is unchanged and native Fluo loses 1.05/1.32 points.

4.3. Effects of tracking components and training data

Adding joint daughter scores produces the largest architectural gain (Table 2). Retraining with zeroed image embeddings isolates appearance. In RG, image features add 0.64–1.20 F1 points on PSC and sparse Fluo (eight-test Holm over R/RG, $p \leq .015$); native Fluo shows no clear effect. Without parent scores, RG image features add 3.16/1.83 sparse points on PSC/Fluo (separate eight-test Holm $p < .01$). Parent-only R scores below the graph reference; parent removal does not significantly change RG F1 after correction. Real-only Pair matches full RG in rounded sparse F1 (Table 1) and is used in matched and transfer tests.

With the full architecture fixed, synthetic data add 1.70 native PSC F1 points, 0.79 sparse PSC points and 0.99 sparse Fluo points; all paired intervals exclude zero. Sparse CT gains from synthetic data are 0.43 points on PSC and 0.76 on Fluo. On native Fluo, R leads RG by 0.39 F1 points ([0.12,0.65]).

4.4. Controls for synthetic supervision

Labels and appearance. Correcting labels on the same rendered images (Raw to RG, Table 3) adds 0.94/0.35 PSC F1 points at 1×/8×. Paired intervals exclude zero on PSC but include zero on Fluo. RG–PM intervals include zero: rendering has no established advantage over painting M3-corrected labels.

Direct-program control. Painting all 200 M1 programs with original labels bypasses generation and M3 (Table 3, lower block). Intensity, texture and halos are fitted on real training frames. Both arms share schedules, 50:50 real/synthetic sampling, 6000 updates and five seeds. Event counts and negative pools are matched per gap, with one positive view per event. Rendering plus M3 adds 0.44 native PSC F1 points (paired 95% interval [0.27,0.62]; four-test Holm

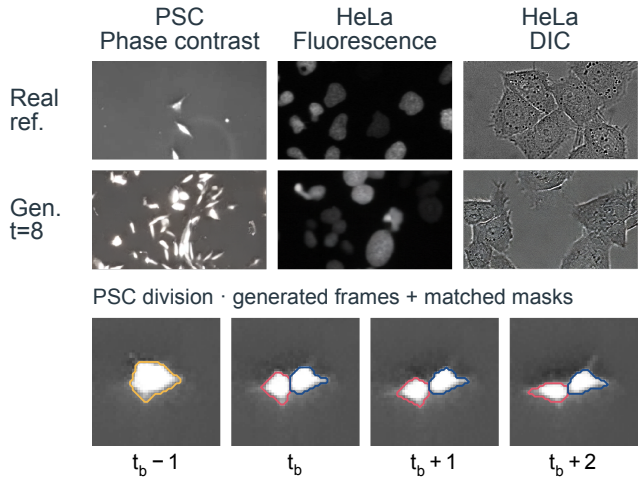


Fig. 2. Real appearance references and generated images; below, consecutive 48×48-pixel PSC crops. References do not fix layout or density. DIC (differential interference contrast) is qualitative; tracking uses PSC/Fluo.

Table 3. Five-seed division F1; bold marks block maxima. Raw: rendered images/program labels; RG/PM: rendered/painted images with corrected labels. Lower block: original-label painting versus rendering/M3; matched events, pair-only decoding.

	PSC 01		Fluo 01	
Data	1×	8×	1×	8×
Full tracker; original sampling				
Raw	.869±.006	.745±.005	.948±.005	.908±.006
RG	.879±.003	.749±.005	**.950**±.002	**.909**±.006
PM	**.885**±.005	**.750**±.004	**.950**±.002	**.909**±.007
Pair-only; matched events				
Direct paint	.884±.006	**.754**±.002	.946±.007	**.916**±.005
Render+M3	**.888**±.006	**.754**±.003	**.950**±.002	.913±.006

$p = .009$); other intervals include zero. Neither arm differs significantly from real-only under separate eight-test Holm correction.

Matched generator and tracker comparisons. Matching events, negative pools and updates, MitoSync exceeds SCF-adapted by 1.19/2.37 sparse PSC/Fluo points (four-test Holm $p < .002$). Native gains (+0.44/+0.20) are not significant; generation costs differ. MAGIK loses 11.6 sparse Fluo points with uniform window sampling (Table 1). Balanced source/gap sampling (18,750 updates per arm) gives real-only/augmented Fluo F1 of .680/.755 at 1× and .616/.630 at 8×; both three-seed intervals include zero.

Dynamics and phase scheduling. With cycle/shape fixed, axis bias, sister constraints, exclusion and entry change F1 by +0.37/+0.12 points on 40 paired PSC programs (1×/8×; two-test Holm $p \geq .058$). A single-phase timing study raises on-time detection by 8.58 points on 24 PSC scenes (block-bootstrap 95% interval [3.95,14.57]). The production two-phase policy (40 programs per dataset) raises first-attempt on-time matches from 140/1283 to 196/1310 on PSC and 295/868 to 348/872 on Fluo. With M3 and training budgets matched, tracking changes are +0.39/−0.16 points on PSC and −0.21/+0.24 on Fluo (1×/8×); none is significant after four-test Holm correction. Phase shifts may change later geometry.

Blinded author review. One author, blinded to M3 decisions, reviewed 72 PSC events from 67 clips, eight per density/decision stratum. Division was visible in 23/24 kept, 16/24 re-timed and 10/24 dropped cases. Weighted rates are 95.9%, 66.5% and 41.2%; the retained rate is 75.6% (approximate clip-clustered 95% interval [61.6,88.0]), including uncertain/lost cases. The weighted absolute onset difference is 1.32 frames (39 retained positives): human review marks first separation; M3 requires two-frame persistence. Clicks match training-ID triplets in 17/39 retained positives. Daughter clicks precede label birth in 20 cases; mother clicks occur at/after label birth in two. Both daughter clicks carry the mother ID in 17 retained and 9/10 dropped positives; these timing disagreements do not measure ancestry error. In eight preselected timing conflicts (mother at frame 0, daughters at 16), 7/8 triplets match training IDs; one maps both daughters to one ID. These checks do not establish corpus accuracy or inter-rater agreement.

Uncertain supervision. Excluding triplets from re-timed or dropped families and descendants changes F1 by −0.31/−0.26 PSC and +0.09/−0.15 Fluo points versus count-matched random subsampling (1×/8×; five seeds; Holm $p \geq .374$). Masking also removes valid examples and shows no clear benefit.

4.5. Frozen-model transfer to new videos

Frozen Fluo models transfer without tuning to 36 LFCT fields (four cell lines [23]). Of 53 fields, exclusions comprise 11 multi-parent cases, four frame-count mismatches, one without signal and one unavailable. Event F1 uses exact triplets and centroid-in-mask assignments on shared Cellpose-SAM detections; it differs from CTC BC(0). Graph/R/RG means are .140/.130/.126 at 1× and .098/.138/.141 at 8×, averaging offsets, seeds, then fields. R/RG sparse gains over graph are 4.05/4.33 points (four-test Bonferroni field-bootstrap intervals [2.38,5.25]/[2.90,5.83]); native intervals include zero. RG minus R is −0.34/+0.28 points; adjusted intervals include zero. Matched rendering/M3 trails painting by 0.41/0.54 points (separate four-test intervals exclude zero). All four cell lines gain sparse R–graph F1 (2.24–6.04 points); native signs vary. Intervals describe this cohort; biological independence is unknown.

With fixed R scores and no continuity, ILP exceeds greedy under the same utility by 0.45/1.45 points (1×/8×; six-test Bonferroni intervals [0.21,0.75]/[0.82,2.17]). Continuity changes sparse R/RG F1 by −0.25/−0.29 points (adjusted intervals below zero); native intervals include zero. This is not a replication of [13].

Computational cost. PSC/Fluo job times sum to 5.41/5.40 hours for LoRA, 5.12/3.97 for generation with retries and 2.35/1.96 for readout, excluding label-editing overhead. Painting 200 clips takes 7.49/7.69 CPU minutes. Concurrent generation used 96 GB RTX PRO 6000 GPUs; incomplete hardware/concurrency records preclude GPU-hour or end-to-end cost ratios.

5. DISCUSSION AND CONCLUSION

Joint daughter scoring and decoding provide the main tracking gains. Ablations and fixed-score controls support both components. LFCT gains support transfer under sparse sampling.

Programmable supervision benefits several fixed-tracker settings, with no consistent advantage over simpler supervision in matched controls. Code and evaluation data: `github.com/CHEN-Yunshu/MitoSync-ISBI2027`.

6. COMPLIANCE WITH ETHICAL STANDARDS

We used public Cell Tracking Challenge [21] and LFCT [23] data. No new human or animal data were collected, so ethical approval was not required. The authors have no relevant financial or non-financial interests to disclose.

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